

Colby Miller

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Education

B.S. Mechanical Engineering, California State University, Long Beach (CSULB) *Expected Fall 2026*
Engineering GPA: 3.51 | Coursework: Fluid Mechanics, Heat Transfer, Thermodynamics I & II, Manufacturing, CAD/CAM, Mechanics of Materials, Materials Science, Design of Machinery, Intro to Control Systems

President's Dean's List — Fall 2025

A.S. Mathematics for Transfer, Cerro Coso Community College *Graduated 2024*

Math Student of the Year (2023–2024) – Awarded by the Math Department for outstanding performance

Experience

Fluids Lead & Propulsion Engineer — Schedule 1 Rocket Project *May 2026 – Present*

- Developed a MATLAB engine performance optimization tool using NASA CEA to evaluate propellant combinations, optimize O/F ratio and chamber pressure, and predict thrust, specific impulse, combustion temperature, and nozzle expansion performance
- Designed a conical converging-diverging nozzle using *Modern Engineering for Design of Liquid-Propellant Rocket Engines*, modeled the geometry in Onshape, and validated performance through species-based ANSYS Fluent CFD using NASA CEA combustion-product mass fractions, achieving less than 1% error in predicted thrust and exit Mach number.
- Coupled Bartz heat-transfer calculations with transient ANSYS Mechanical thermal analysis to predict combustion-chamber wall temperatures and guide material selection for a heat-sink copper rocket engine
- Designed the complete LOX/ethanol feed system including P&IDs, component selection, pressure-loss analysis, instrumentation layout, valve/regulator sizing, and full 3D CAD implementation in Onshape

Assistant Fluids Lead, Beach Launch Team — California State University, Long Beach *May 2025 – Present*

- Supported manufacturing and integration of the rocket propulsion fluids system from initial CAD designs through final assembly, including tube cutting, bending, AN flaring, lower fluids mounting hardware fabrication, and machining pneumatic actuator arms for the main valves
- Directed a 12-run water-flow test campaign to measure feed-system flow rate, pressure drop, and flow coefficient (Cv), analyzing experimental results to establish the required propellant tank pressures for Theseus static fire and flight operations.
- Executed cryogenic LOX cold-flow testing to validate plumbing, valve operation, and integrated propulsion-system performance while identifying and resolving hardware issues prior to static fire
- Conducted high-pressure leak and proof testing of the propulsion system at pressures up to 3,750 psi to verify system integrity prior to cryogenic operations
- Supported integrated static-fire testing and Beach Launch Team's first liquid rocket launch in six years by performing LOX loading, propulsion-system troubleshooting, leak verification, and ground-support fluid operations

Applications Engineering Co-op — Flowserve Corporation *May 2026 – Present*

- Support mechanical seal failure analyses for industrial rotating equipment by inspecting returned seals, documenting wear patterns, disassembling components, and identifying potential failure mechanisms for customer engineering reports
- Evaluate operating conditions using maintenance history, seal piping plans, P&IDs, pump performance curves, vibration data, and field inspection measurements to assist in determining equipment root cause and corrective actions
- Developed a Python automation tool using pandas to process SAP bill-of-materials data and company equipment databases, automatically matching seal kits to installed pump configurations and generating upload-ready Excel files for enterprise database integration, eliminating a previously manual engineering workflow involving over \$500,000 of unmanaged inventory

Software and Skills

- **Software:** SolidWorks, Onshape, Siemens NX, ANSYS Mechanical, ANSYS Fluent, MATLAB, Python, Fusion 360 (CAM), Microsoft Excel
- **Fluid Systems & Testing:** High-pressure (3000+ psi) flow testing, cryogenic flow testing, pressure decay testing, fluid system design, P&ID development, component selection and sizing
- **Fluid Hardware:** Dome-loaded and reducing regulators, solenoid valves, ball valves, check valves, fittings (NPT, AN, Swagelok)
- **Instrumentation & Data Acquisition:** Pressure transducers, LabJack T7 Pro (LJM), thermocouples, system wiring, sensor integration
- **Thermal & Engineering Analysis:** Root Cause Analysis, Pressure drop analysis (Darcy–Weisbach, minor losses), thermal analysis, MATLAB and Excel modeling
- **Manufacturing & Fabrication:** Manual machining (lathe, mill, drill press), CNC machining, tube bending and flaring, fittings integration, 3D printing, GD&T